Lesson notes · for you

Integer programs and branch and bound

Optimization · Level 4 · Lesson 2

Time	Materials
25 min	Your phone, or a computer with Colab; Paper for drawing a branch-and-bound tree

Learning goals

- Explain why rounding an LP can fail, and why the relaxation is a bound.
- Follow branch and bound: branch, bound, prune, and the gap.
- Model whole units and a fixed cost in SCIP.

Activity: Activity: The bus tree by hand

1. Root (5 min): The cheapest cost per seat is the bus (550 ÷ 30 ≈ 18.3 against 300 ÷ 14 ≈ 21.4), so the relaxation uses buses only: 64 ÷ 30 = 2.13 buses, 1,173.33 coins. Write it at the top of a tree.
2. Branch (8 min): Draw two children: “at most 2 buses” and “at least 3 buses”. In the first, 2 buses seat 60, and the relaxation adds 4 ÷ 14 ≈ 0.29 of a minibus: about 1,186. In the second, 3 buses cost 1,650.
3. Prune (7 min): Keep branching the first child on the minibus (at least 1). You reach 2 buses and 1 minibus, 1,400, a whole plan. The 3-bus branch can’t beat it (1,650), so cross it out without exploring it.
4. Fee (5 min): Add a 300-coin booking fee for any minibus. Recompute: 2 buses + 1 minibus = 1,700, so 3 buses (1,650) win. At what fee does the answer flip? (250.)

Work in pairs: one draws the tree, the other checks each relaxation. Then run the same model in the lesson’s code step and compare.

Discussion

- Why can rounding an LP answer break a constraint?
- Why does a smaller big-M make branch and bound faster?
- When is a 3% gap good enough, and who decides?

Check yourself

- A relaxation costs 900 and your best plan 1,000. Write the gap, and what it promises.

Try it this week

Model the field trip in SCIP with the 300-coin booking fee (1,650). Then change the fee and find where the answer flips from three buses back to two buses and a minibus. Explain the flip point in one sentence.

Worksheet

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Name _____ Class _____ Date _____

1 **64 people; a minibus seats 14 for 300 coins and a bus 30 for 550. Without fees the best plan is 2 buses and 1 minibus (1,400). The minibus company now charges a 300-coin booking fee if you hire any minibus. What is the best plan now?**

◯ Still 2 buses and 1 minibus (1,400)
◯ Three buses, 1,650: with the fee the old plan costs 1,700
◯ Hire five minibuses instead

2 **Put the lines of a branch-and-bound search for a knapsack in order. bound(i, room) is an optimistic estimate of what loads i onwards could still add. Explore taking a load before skipping it.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `best = min(best, value)`
☐ `return`
☐ `def search(i, room, value):`
☐ `w, v = items[i]`
☐ `if i == n or value + bound(i, room) <= best:`
☐ `best = max(best, value)`
☐ `search(i + 1, room, value)`
☐ `global best`
☐ `search(i + 1, room - w, value + v)`
☐ `if w <= room:`
☐ `search(i + 1, room + w, value - v)`

3 **In a minimisation problem, the LP relaxation's cost can never be higher than the cost of the best whole-number plan.**

◯ True ◯ False

4 **The minibus company now charges a 300-coin booking fee if you hire any minibus. Add a yes/no variable `book`, a limit so minibuses need a booking (at most 10 when booked), and the fee in the cost. What does the trip cost now?**

```
from ortools.linear_solver import pywraplp
s = pywraplp.Solver.CreateSolver("SCIP")
small = s.IntVar(0, 10, "minibuses")
big = s.IntVar(0, 10, "buses")
s.Add(14 * small + 30 * big >= 64)
s.Minimize(300 * small + 550 * big)
s.Solve()
print(int(s.Objective().Value()))
```

Write your program:

5 **Which of these decisions need whole-number or yes/no variables? Choose all that apply.**

- [ ] How many buses to hire
- [ ] Whether to open a new tutoring centre
- [ ] Kilograms of rice to buy for the canteen
- [ ] Which lecturer invigilates which exam
- [ ] Litres of water for the garden each week

✎ **Where have you seen a decision rounded when it should have been solved with whole numbers? What went wrong, or might have?**